

Session 1 Speaker Script

Introduction to AI, Generative AI, and Agentic AI

Presenter	Dinesh Wadhwani
Session	1 of 12
Total duration	≈ 60 minutes (12 slides, detailed talk track for each)
Purpose	Full spoken script — roughly 5 minutes of talking content per slide

How to use this document

This companion script is designed to be read alongside the Session 1 slide deck. For each slide, you'll find: what's on screen (a quick recap), a detailed word-for-word style speaking script written to fill roughly five minutes of natural, conversational delivery, and a short list of delivery tips — pacing notes, discussion prompts, and moments to pause. You don't have to read the script verbatim; it's written so you can absorb the flow and deliver it in your own words.

At a Glance: Slide-by-Slide Timing

#	Slide Title	Approx. Time
1	Introduction to AI, Generative AI, and Agentic AI	~3 min
2	How We'll Spend the Next 60 Minutes	~2 min
3	A Brief History of AI	~8 min
4	From Traditional AI to Agentic AI	~10 min
5	Predictive AI vs Generative AI vs Agentic AI	~7 min
6	GenAI & LLM Fundamentals — At a Glance	~5 min
7	What Makes a System “Agentic”?	~6 min
8	Chatbot vs Copilot vs Assistant vs Agent	~6 min

9	Real-World Examples You Already Use	~5 min
10	Classroom Exercise: Generate a Resume with GenAI	~12 min
11	Key Takeaways from Session 1	~3 min
12	Session 2: LLM Fundamentals and Output Behavior	~2 min

Timings are approximate and total slightly above 60 minutes to allow for natural classroom interaction, questions, and the lab. If you need to trim time, compress Slide 3 (History) or Slide 9 (Examples) first — protect Slide 4 (the evolution ladder) and Slide 10 (the lab).

Introduction to AI, Generative AI, and Agentic AI

~3 min

WHAT'S ON SCREEN

- Course name, Session 1 of 12
- Presenter name
- Session focus line and duration (60 minutes)

SPEAKING SCRIPT — What to Say

Good morning/afternoon everyone, and welcome to the first session of our course on Autonomous AI Systems and Agent-Based Computing. My name is Dinesh Wadhvani, and over the next twelve sessions we're going to build up, brick by brick, from 'what is AI, really' all the way to designing multi-agent systems that can plan, use tools, and act with real autonomy.

Before I dive in, let me set expectations for how this course runs. Every single session — all twelve of them — has a hands-on lab component. This isn't a course where you just listen to slides and take notes; you will build something every session, even if it's small. AI is a hands-on discipline, and the fastest way to actually understand these concepts is to use the tools yourself.

Here's something fun: today, in this very first session, you're going to create a fictional student's resume using a GenAI tool. That might sound like a strange way to start an AI course — but that resume becomes the running example for the entire course. In Session 2 you'll test different prompts on it. In Sessions 4 and 5 you'll refine and restructure it. In Sessions 7–9 you'll compare it against job descriptions using embeddings and retrieval. And by Sessions 10–11 you'll build actual AI agents that review, tailor, and finalize that same resume automatically. What feels like a small warm-up today is the seed of a project that grows across the whole course.

Today, Session 1, is the foundation everything else stands on. We'll cover: the history of AI and how it evolved into what we have today; the different types of AI you'll hear about constantly — predictive, generative, agentic; the basic vocabulary around GenAI and large language models; and the genuinely confusing difference between a chatbot, a copilot, an assistant, and an agent. By the end of the hour you should be able to look at any AI product and correctly categorize what kind of system it actually is.

Quick note on pace: today is intentionally concept-heavy and light on code — we're building your mental model first. Starting next session, things get progressively more hands-on and technical. Use today to ask questions and build intuition. Alright — let's get started.

DELIVERY TIPS

- *Keep this to 3 minutes; don't over-explain the whole course yet — just enough to set the hook.*
- *Good moment for a quick show of hands: "Who has used ChatGPT before?" to gauge the room.*
- *A 10-second personal note on why you're excited about agentic AI lands well here.*

02

SESSION ROADMAP

How We'll Spend the Next 60 Minutes

~2 min

WHAT'S ON SCREEN

- 8-item agenda with time allocations
- History → Evolution ladder → Types of AI → GenAI/LLM basics → Agentic traits → Terminology → Lab → Wrap-up

SPEAKING SCRIPT — What to Say

Let me quickly walk you through how we'll spend our hour together, so you always know where we are in the session.

We'll start with a brief history of AI — about eight minutes — tracing how we got from 1950s rule-based systems to the agentic AI systems making headlines today. Then we'll spend about ten minutes on what I think is the single most important slide of the day: the evolution ladder from Traditional AI, to Machine Learning, to Deep Learning, to Generative AI, and finally to Agentic AI. If you remember only one thing from today, I want it to be that ladder.

After that, about eight minutes distinguishing three types of AI you'll hear constantly: predictive, generative, and agentic. Then a shorter six-minute segment on GenAI and LLM fundamentals — just enough vocabulary to get comfortable, since we go much deeper in Session 2.

We'll spend six minutes on what actually makes a system 'agentic' — a preview of Session 10 — and another six minutes clearing up genuinely confusing terminology: chatbot versus copilot versus assistant versus agent, terms even professionals mix up.

Then the fun part: roughly twelve minutes hands-on, where you'll actually use a GenAI tool to build a resume for a fictional student. We'll close with a short wrap-up covering key takeaways and a preview of Session 2.

Notice the pattern, because it repeats every session in this course: concept, then example, then hands-on lab. If we run long on discussion earlier, I will compress the history section before I compress your lab time — that hands-on time is protected.

DELIVERY TIPS

- *Move briskly — students don't need every minute justified, just the shape of the hour.*
- *Natural moment to let latecomers settle before the dense content starts.*

03

FOUNDATIONS

A Brief History of AI

~8 min

WHAT'S ON SCREEN

- Timeline: 1950s Birth of AI → 1980s Expert Systems → 1990s–2000s Machine Learning → 2012–2019 Deep Learning → 2020–2022 Generative AI → 2023–Now Agentic AI

SPEAKING SCRIPT — What to Say

Let's start at the beginning. I want you to leave this slide understanding that ChatGPT did not appear out of nowhere in 2022 — AI has been through decades of cycles, including periods we call 'AI winters,' where funding and interest collapsed because the technology couldn't live up to its promises. Let's go decade by decade.

The 1950s — Birth of AI. In 1950, Alan Turing published a paper asking a deceptively simple question: 'Can machines think?' He proposed what we now call the Turing Test — if a human evaluator can't reliably distinguish a machine's conversation from a human's, the machine passes. Then in 1956, at a workshop at Dartmouth College, researchers coined the actual term 'Artificial Intelligence.' That's the official birthplace of the field as a named discipline — almost entirely theoretical at this point, since the hardware to do anything meaningful didn't exist yet.

The 1980s — Expert Systems. This era tried to encode human expertise directly into software using rules — thousands of hand-written 'if this, then that' statements. A medical diagnosis system might have a rule like: if the patient has a fever AND a rash AND recently travelled to a specific region, THEN suspect a specific disease. Genuinely useful in narrow domains, but brittle — if a situation fell even slightly outside the anticipated rules, the system had no way to adapt. Maintaining thousands of hand-written rules became its own nightmare, and that's a big reason for the next AI winter.

The 1990s into the 2000s — Machine Learning Rises. Instead of a human writing the rules, what if the system learned the rules itself from examples? That's the fundamental shift: statistical learning from data instead of hand-coded logic. Think of early email spam filters: rather than a person writing 'if the email contains a specific word, mark as spam' — which spammers trivially evade — the system learns statistical patterns from thousands of labeled spam and non-spam emails.

2012 through 2019 — the Deep Learning Boom. A specific, very powerful sub-family of machine learning based on neural networks, loosely inspired by how neurons connect in the brain. In 2012, a neural network called

AlexNet dramatically outperformed every other approach in an image recognition competition — often cited as the spark for this era. What made this possible now, and not decades earlier when neural networks were first theorized? Two things: massive internet-scale datasets, and GPUs — graphics chips built for video games that turned out to be extremely good at exactly the math neural networks need.

2020 to 2022 — Generative AI Arrives. Models like GPT-3 and image generators like DALL-E showed these deep learning models could not just classify or predict — they could create genuinely new content: essays, poetry, images from a text description. This is when the general public really started noticing AI in a new way.

2023 to now — Agentic AI Emerges. The newest shift, and the whole reason this course exists. Instead of a model just responding to one prompt, these systems can plan a sequence of steps, decide to use external tools, observe what happens, and adjust — with much less step-by-step human instruction.

Discussion prompt for the room: where do you think we are on this curve right now, today? And more interestingly — what comes after agentic AI? I don't expect a right answer; I want you thinking about the trajectory, not just the history.

DELIVERY TIPS

- *This is content-dense — use the visual timeline as your anchor and expand verbally rather than reading the slide.*
- *The 'AI winter' concept is a great hook — it recontextualizes the assumption that AI progress is a smooth, inevitable line.*
- *If short on time, compress the 1980s and 1990s–2000s beats into one combined point.*

04

THE BIG PICTURE

From Traditional AI to Agentic AI

~10 min

WHAT'S ON SCREEN

- 5-stage ladder: Traditional AI → Machine Learning → Deep Learning → Generative AI → Agentic AI
- Each stage adds a capability; none replace the last

SPEAKING SCRIPT — What to Say

This is, in my opinion, the single most important slide of today's session. If you forget everything else, remember this ladder — and notice I called it a ladder, not a replacement chain. Each rung adds a new capability on top of the one before; it doesn't throw away the previous rung.

Traditional AI is the starting point: entirely rule-based, if-then logic, hand-coded by a human engineer. A basic chess-playing program from decades ago, or a tax calculator. Zero learning happens — if you don't update the code, the behavior never changes, no matter how much data flows through it.

Machine Learning changes the fundamental approach. Instead of a person writing the rules, we give the system labeled examples, and it learns the underlying pattern itself. Credit scoring is a good example — instead of a human writing rules for creditworthiness, the system learns from historical data on thousands of past loans and their outcomes. The behavior comes from data, not hand-written logic.

Deep Learning is a specific, very powerful family within machine learning, using layered neural networks. Why does this matter separately from 'machine learning' in general? Because deep learning is what made image recognition, speech recognition, and eventually language understanding practical at massive scale — which is exactly why it only took off once GPUs and internet-scale data became available.

Generative AI is a deep learning model with a specific purpose: instead of just classifying something or predicting a number, it's trained to produce new content — text, images, audio, video, code. This is the category ChatGPT and Midjourney belong to.

And finally, Agentic AI — the newest and most advanced rung. A system that doesn't just respond to one prompt; it can plan a multi-step approach to a goal, decide which tool to use, take actions, observe results, and adjust its plan — with meaningfully less moment-to-moment human instruction than any previous stage.

Here's the critical insight I want you to sit with: agentic AI today isn't a totally separate technology from everything before it — it's built ON TOP of deep learning and generative models. The 'brain' making decisions inside an agent is very often a large language model, which is a generative AI system. This ladder isn't five unrelated technologies; it's five layers of increasing capability, each standing on the shoulders of the last.

Quick understanding check — I'll describe a system, you tell me which rung it belongs to. A spam filter that learns from labeled emails? [Machine Learning] A tool that writes a poem about your dog from a text description? [Generative AI] A thermostat that turns on the heat below 18 degrees because an engineer coded that threshold? [Traditional AI] A system that books your entire vacation — researching, comparing, and completing the booking — from just a starting request? [Agentic AI]

DELIVERY TIPS

- *Spend real time here — close to 10 minutes given how central it is to the whole course.*
- *The mini-quiz works well as a cold-call or show-of-hands; use it to confirm the room absorbed the ladder before moving on.*
- *If tight on time elsewhere, borrow it from Slide 3 or Slide 9, never from this slide.*

Predictive AI vs Generative AI vs Agentic AI

~7 min

WHAT'S ON SCREEN

- Predictive AI — “What will happen?”
- Generative AI — “Create something new.”
- Agentic AI — “Achieve this goal for me.”

SPEAKING SCRIPT — What to Say

We just looked at AI as an evolutionary ladder. Now let's slice it a different way — by what kind of task the system is built to handle. This gives us three practical categories: predictive, generative, and agentic.

Predictive AI answers ‘what will happen?’ It looks at historical data, finds patterns, and outputs a score, a label, or a forecast — it does not create anything new. Fraud detection is a classic example: a bank's system looks at a transaction and outputs a probability score — how likely is this specific transaction to be fraudulent? Demand forecasting is another: predicting how many units a retailer will sell next month based on historical sales. Churn prediction — will this customer cancel their subscription in the next 30 days? All grounded in patterns from the past.

Generative AI flips the task completely: instead of predicting a value, it creates something new. Given a prompt, it produces original text, images, audio, video, or code that didn't exist before you asked for it. ChatGPT writing an essay, Midjourney generating an image from a description, GitHub Copilot suggesting a whole function of code — the output is a new artifact, not a forecast.

Agentic AI answers a different question entirely: ‘achieve this goal for me.’ Rather than one input producing one output, an agent is given an objective, plans a sequence of steps toward it, potentially uses tools along the way, and acts with real autonomy. An AI travel-booking agent doesn't just suggest flights — it can search across sites, compare prices, and complete an actual booking. An autonomous coding agent doesn't just suggest one line of code — it can read a codebase, make a plan, write multiple files, run tests, and open a pull request.

Simple mental shortcut: predictive AI answers a question. Generative AI creates an artifact. Agentic AI completes a task.

Important nuance: these categories aren't rigid boxes — a lot of real systems blend them, and a single system can evolve from one category to another over time. Remember the resume exercise you're about to do today? Right now you're using a purely generative tool — you give it a prompt, it creates a resume. But by Session 10, we'll build an actual agent that reviews that same resume against a job posting and autonomously suggests — or makes — improvements. Same underlying task, evolving from generative to agentic as we add planning and autonomy on top.

DELIVERY TIPS

- Use the resume callback deliberately — it reinforces the course throughline and previews later sessions.
- If time is short, the fraud-detection and coding-agent examples are the two most vivid — you can drop one of the others.

CORE CONCEPTS

06

GenAI & LLM Fundamentals — At a Glance

~5 min

WHAT'S ON SCREEN

- Generative AI (GenAI) — umbrella term for content-creating models
- Large Language Model (LLM) — GenAI focused on language
- How it works in one line: next-token prediction
- Why it matters: the reasoning engine inside GenAI apps and future agents

SPEAKING SCRIPT — What to Say

Let's nail down two terms you'll hear constantly for the rest of this course, and honestly, for the rest of your career if you work anywhere near AI: GenAI and LLM.

Generative AI, or GenAI, is the umbrella term — the category of AI models trained to produce new content rather than just classify or score data, as we just discussed. It covers text generators, image generators, music generators, code generators — the whole family.

A Large Language Model, or LLM, is a specific and very important member of that family — one focused specifically on language. It's a model trained on truly enormous amounts of text — books, websites, articles, code — to understand and generate human language. GPT, which powers ChatGPT, Claude from Anthropic, Gemini from Google, Llama from Meta — these are all LLMs.

Here's the one-line explanation of how an LLM works that I want you to hold onto, because it will demystify a lot of confusing behavior later in this course: an LLM predicts the next most likely word — or more precisely, the next 'token' — one at a time, based on everything that came before it in the prompt and its own response so far. It does this over and over until it produces a full response.

Why does this matter so much? Because it explains something that confuses a lot of newcomers: LLMs can sound extremely confident while being factually wrong. The model isn't retrieving verified facts from a database — it's predicting plausible, statistically likely text based on patterns learned during training. Usually that produces correct, useful answers, because correct information tends to be the statistically common pattern in huge amounts of text — but not always, and that gap is exactly why techniques like Retrieval-Augmented

Generation, which we cover in Sessions 8 and 9, exist: to ground the model's output in actual retrieved documents.

Why does this matter for where this course is going? Because the LLM is the 'reasoning engine' inside almost every GenAI application — and it's also the reasoning engine inside the autonomous agents we build toward later in this course. When an agent 'plans' or 'decides,' under the hood, very often, it's an LLM doing exactly what I just described: predicting the next best token, now applied to planning steps and tool choices instead of just conversational text.

We'll go much deeper on tokens, transformer architecture, pretraining versus fine-tuning, and context windows in Session 2 — today was just enough to get the vocabulary comfortable in your mouth.

DELIVERY TIPS

- *This slide is intentionally light — resist going deep into transformer architecture today; that's next session's job.*
- *The 'confident but wrong' point is worth a pause — ask if anyone has personally seen an LLM confidently state something false.*

07

DEEPER LOOK

What Makes a System “Agentic”?

~6 min

WHAT'S ON SCREEN

- Reasoning · Planning · Memory · Tool Use / Action
- “A GenAI model becomes an agent the moment it can plan, remember, and act.”

SPEAKING SCRIPT — What to Say

We've used the word 'agentic' a lot already today, so let's actually define it. What specific capabilities turn a generative model into something we'd call an agent? Four core traits worth knowing today — we go into real depth on each in Session 10, so treat this as a preview.

Reasoning: the system breaks down a goal and thinks through how to approach it, rather than reacting to a single instruction in isolation. It's asking itself, in effect, 'what does this goal actually require?'

Planning: once it's reasoned about the goal, it creates an actual sequence of steps to get there — not one action, a plan, potentially with several ordered steps.

Memory: the system retains context across those steps. This matters enormously for multi-step tasks — imagine trying to book a trip if you forgot, halfway through, what dates you'd already searched. Some agents keep memory just within a single task; more advanced ones retain memory across entire sessions.

Tool Use, or Action: the system can call external tools, functions, or APIs to actually act in the world, rather than just generating a text description of what someone else should do. That's the difference between an agent saying 'here's how you'd search for flights' and an agent actually querying a flight-search API itself.

Concrete before-and-after example: you ask an AI system, 'Book me the cheapest flight to Mumbai next weekend.' A plain chatbot — no agentic capability — responds with a description: 'Here are some steps you could take: check a few sites, compare airlines...' It hands the work back to you. An actual agent searches real flight data across sources, compares actual prices, and either completes the booking directly or presents a specific, ready-to-approve choice — 'I found a flight for eight thousand two hundred rupees on IndiGo, departing Saturday at 6 AM — shall I book it?' That's the qualitative difference reasoning, planning, memory, and tool use together produce.

One sentence I want you to leave with, the cleanest definition of 'agentic' you'll find: a GenAI model becomes an agent the moment it can plan, remember, and act — not just respond once. We build directly on that sentence starting in Session 10.

DELIVERY TIPS

- *Keep the flight-booking example vivid and specific with actual numbers — concreteness sticks better than abstraction.*
- *Good moment for a quick poll: "Who has already used an AI agent without realizing it was 'agentic'?"*

08

TERMINOLOGY CLINIC

Chatbot vs Copilot vs Assistant vs Agent

~6 min

WHAT'S ON SCREEN

- Chatbot — answers / converses, reactive, no autonomy
- Copilot — assists inside a specific tool, human stays in control
- Assistant — manages tasks across contexts, still mostly reactive
- Agent — pursues a goal autonomously, plans and acts

SPEAKING SCRIPT — What to Say

This slide clears up something that genuinely confuses even people working in the industry, because these four words get thrown around so loosely: chatbot, copilot, assistant, and agent. Let's define each clearly, then I'll show you they sit on one continuous spectrum.

A Chatbot answers questions or holds a conversation. Think of the customer-support chat widget on an e-commerce site. It's reactive — it responds to what you say — and largely focused on a single turn or short back-and-forth, with essentially no autonomy.

A Copilot works alongside you, embedded inside a specific tool, to help you do your job faster. GitHub Copilot is the namesake example — suggesting code inline as you type in your editor. The human stays firmly in the driver's seat; the AI assists and suggests, but you decide what to accept.

An Assistant has a broader scope than a copilot — it helps manage tasks across multiple contexts rather than being embedded in one tool. Siri, Alexa, Google Assistant — calendar, reminders, quick questions, smart-home controls. Still largely reactive to your requests, but spanning more of your life than a copilot.

An Agent is the most autonomous of the four. Given a goal, it can plan multiple steps, invoke tools, and carry out actions with meaningfully reduced supervision — checking in with a human only at key decision points, rather than every single step.

The pattern to see: autonomy increases left to right. Chatbot is almost entirely reactive. Copilot assists but keeps you in the driver's seat. Assistant handles more breadth but is still mostly reactive. Agent takes the wheel itself, checking in with you only when it matters.

Quick interactive check — I'll name a tool, you tell me where you'd place it, and don't worry about being exactly right, there's genuine debate even among practitioners: Google Maps suggesting a route? [Somewhere between assistant and agent, leaning assistant, since it doesn't drive the car itself.] An AI trip planner that books flights and hotels without further confirmation? [Agent.] A customer-support widget answering FAQs? [Chatbot.]

Why this vocabulary matters specifically for you: in the job market and in product descriptions, companies use these words inconsistently, sometimes for marketing purposes — calling something an 'agent' when it's really closer to a chatbot with extra features. Having this spectrum in your head lets you cut through the marketing and evaluate what a system actually does.

DELIVERY TIPS

- *The interactive naming exercise works best live — pick tools your specific class will recognize.*
- *Explicitly flag that marketing language often overstates 'agent' — a useful critical-thinking moment.*

09

SEEING IT IN ACTION

Real-World Examples You Already Use

~5 min

WHAT'S ON SCREEN

- Netflix / Spotify — predictive
- ChatGPT, Copilot, Midjourney — generative
- Siri, Alexa, Google Assistant — assistant
- AI travel & coding agents, autonomous driving — agentic
- Fraud detection at checkout — predictive

SPEAKING SCRIPT — What to Say

Let's make everything we've discussed today concrete, because I guarantee you've personally used at least four or five of the examples on this slide within the last week.

Netflix and Spotify recommendations — predictive AI. It looks at your viewing or listening history and everyone else's patterns, and predicts what you're statistically likely to enjoy next. No new content is being created; it's a prediction about your preferences.

ChatGPT, GitHub Copilot, Midjourney — generative AI. Each takes a prompt and produces genuinely new content: essays, code, images.

Siri, Alexa, Google Assistant — our assistant category. Broader than a single embedded tool, handling a range of everyday voice-driven tasks, still largely reactive.

AI travel-booking agents and autonomous coding agents — agentic AI. These plan, search, and execute multi-step tasks — for example, an autonomous coding agent that reads a bug report, locates the relevant code across a whole repository, writes a fix, runs the tests, and opens a pull request, largely on its own.

Fraud detection at checkout — back to predictive AI. Every time you tap your card, a model scores that specific transaction in real time for fraud risk, based on patterns like location, amount, and your typical spending history.

Autonomous-driving systems — one of the most sophisticated real-world agentic examples that exists today. The car continuously perceives its environment, plans a path, and acts — braking, steering, accelerating — in a live, constantly changing environment, with extremely high stakes.

Notice something across this whole list: most real products actually blend more than one category. A modern car with adaptive cruise control uses predictive AI to estimate distance and speed of the car ahead, and increasingly agentic behavior to act on that in real time. Spotify uses predictive AI for recommendations, but its AI DJ feature uses generative AI for spoken commentary. Don't think of these categories as rigid, mutually exclusive boxes — think of them as capabilities real products often combine.

DELIVERY TIPS

- *Ask the class before revealing each category: "which bucket does this belong to?" — more engaging than stating it yourself.*
- *Swap in a locally relevant product (a regional food-delivery ETA, a local ride-hailing app) for extra relevance if useful.*

Classroom Exercise: Generate a Resume with GenAI

~12 min

WHAT'S ON SCREEN

- Steps: invent profile → open GenAI tool → write prompt → review output → present
- Goal: your first hands-on use of Generative AI
- Deliverable: prompt + resume, reused in Sessions 2–11

SPEAKING SCRIPT — What to Say

Alright — time to actually build something. This is our hands-on lab for today, and I want everyone actively working, not just watching.

Step one: invent a fictional student profile — literally make someone up. Name, degree or branch of study, two or three skills, one internship or project. Don't overthink this — thirty seconds of invention is enough. Example: 'Priya Sharma, final-year Computer Science student, skills in Python, SQL, and basic machine learning, completed a summer internship building a small data dashboard.'

Step two: open any GenAI tool you have access to — ChatGPT, Claude, Gemini, or Copilot, whatever's available and approved for our lab today.

Step three: write a single prompt asking it to generate a one-page resume for that fictional profile. Keep your first attempt simple — don't overengineer the prompt yet. Something like 'Write a one-page resume for a final-year computer science student named Priya Sharma with skills in Python, SQL, and machine learning, who completed a data dashboard internship' is a perfectly good first try.

Step four: review what comes back. Is it complete? Does it look realistic? Is it well-formatted for an actual resume? Don't fix it yet if it's rough — that's completely fine and actually expected. Session 2, next time, is entirely about how changing your prompt changes output quality, so seeing some rough edges today naturally is fine — don't over-correct.

Step five: be ready to present. In a few minutes I'll ask for two or three volunteers to show their prompt and the resume it produced, right here on screen.

While you're working, I'll be walking around — flag me down if your tool isn't responding, if you're not sure how to phrase something, or if you want a sounding board on your fictional profile.

[Give students 8–10 minutes of actual work time here before continuing.]

Okay, let's see a couple of these. [Call for two or three volunteers, project their screen, briefly discuss the prompt and resume quality — about 2 minutes.]

One thing before we move on: hold onto this resume and this exact prompt — save it somewhere, don't lose it. This is not a throwaway exercise. You are going to reuse and refine this same resume in Sessions 2 through 11. It becomes your running example through prompt engineering, structured outputs, API-based generation, embeddings-based job matching, retrieval-augmented question answering, and eventually, actual autonomous agents that review and improve it for you. What you build in the next ten minutes has payoff for the rest of the course.

DELIVERY TIPS

- *This is the protected time block — don't compress it even if you're running behind elsewhere.*
- *Circulate actively; don't just stand at the front during build time.*
- *When picking volunteers to present, prioritize variety — a rough output and a polished one both make for good discussion.*

11

WRAP-UP

Key Takeaways from Session 1

~3 min

WHAT'S ON SCREEN

- 5-point recap of the session's core ideas
- Open floor for questions

SPEAKING SCRIPT — What to Say

Let's bring today together with five things I want you to walk out of this room remembering.

One: AI has evolved in stages — traditional AI, to machine learning, to deep learning, to generative AI, to agentic AI — and each stage adds a new capability on top of the last; none of them get thrown away.

Two: predictive AI forecasts, generative AI creates, and agentic AI acts. Three distinct categories, but real products often blend them.

Three: 'agentic' specifically means planning, reasoning, memory, and tool use, all in pursuit of a goal — not simply answering a single prompt well.

Four: chatbot, copilot, assistant, and agent aren't four unrelated marketing terms — they sit on one continuous spectrum of increasing autonomy.

And five — a little different from the others — you've already generated your first piece of GenAI content today, with that resume exercise. That wasn't just busywork; you'll be building on it, literally, for the rest of this course.

Before we close, let me open the floor: does anyone have a question about anything we covered today — the history, the ladder, the categories, the terminology, or the lab? [Pause genuinely here — don't rush past this.]

[If time permits] Let me put one of you on the spot — [name a student] — in your own words, what's the difference between a copilot and an agent?

Great. That's exactly the kind of understanding I want everyone to leave with today.

DELIVERY TIPS

- *Genuinely pause for questions — resist filling silence immediately; students often need a few seconds before raising a hand.*
- *The cold-call is optional depending on classroom culture — use your judgment.*

COMING UP NEXT

12

Session 2: LLM Fundamentals and Output Behavior

~2 min

WHAT'S ON SCREEN

- Preview: tokens, transformers, pretraining vs fine-tuning, context windows
- Lab preview: re-run today's resume prompt with variations

SPEAKING SCRIPT — What to Say

To close, let me tell you exactly what's coming in Session 2, so you know what to expect.

We established today, conceptually, what LLMs and Generative AI are. Next session, we open the hood and look at how LLMs actually behave under the surface — but I promise, without heavy math. We'll cover tokens and tokenization — why an LLM 'reads' your text the way it does, and why that matters for cost and behavior. We'll look at transformers, the architecture underneath every modern LLM, explained conceptually. We'll cover the difference between pretraining and fine-tuning — why a model can know general things but still need extra training for a specific task. And we'll talk about context windows and parameter settings — why there's a limit to how much text you can feed a model at once, and how settings like temperature change how creative or predictable its output is.

And remember that resume you built today? In Session 2's lab, you'll take that exact same profile and test it with multiple different prompt phrasings, side by side, and compare the output quality. You'll start to see, very concretely, why the exact words you choose in a prompt matter so much — which sets up prompt engineering, which we dive into properly in Sessions 4 and 5.

Thank you all for a great first session — I've enjoyed the questions and energy in the room today. See you next time.

DELIVERY TIPS

- *Keep this brief and warm — it's a closing beat, not a new lecture.*
- *If running over time, this is the one slide you can compress to thirty seconds without losing much.*